Noise Schemas Aid Hearing in Noise

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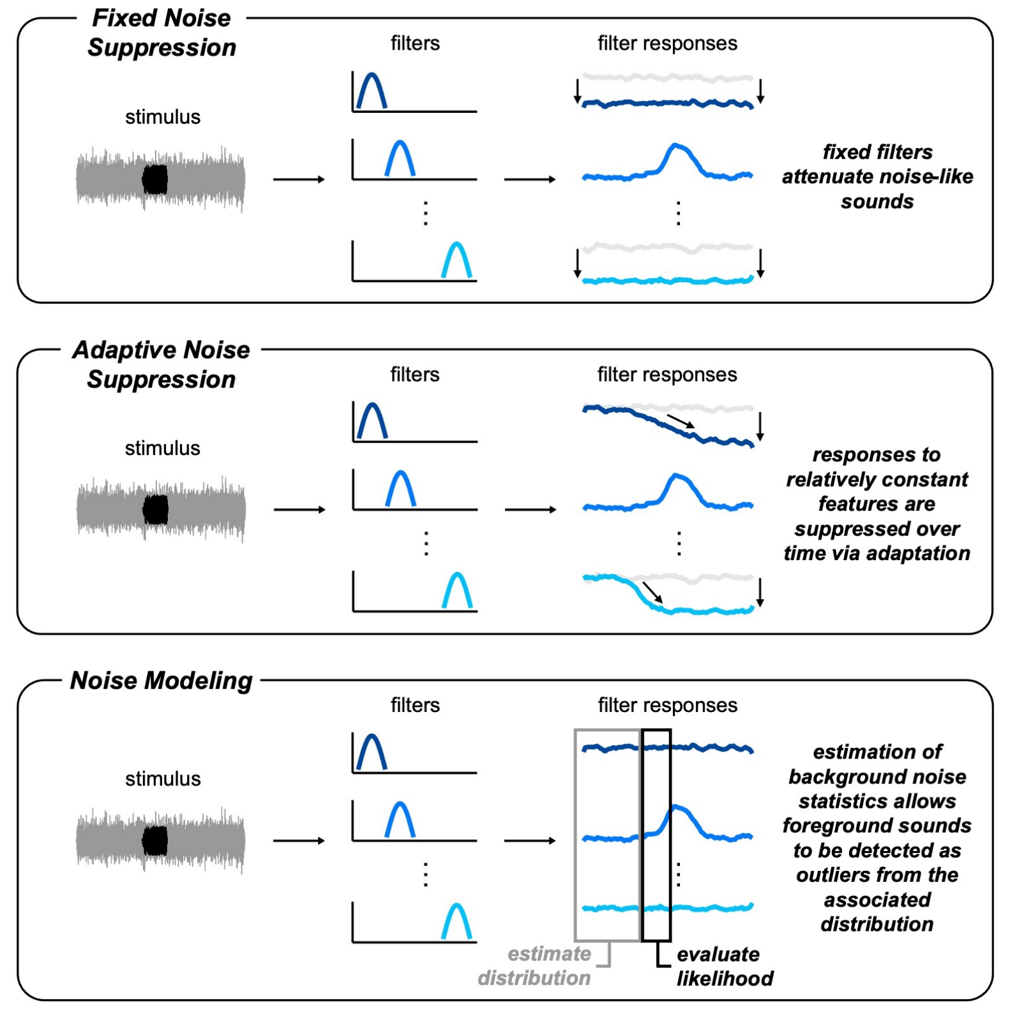
Introduction

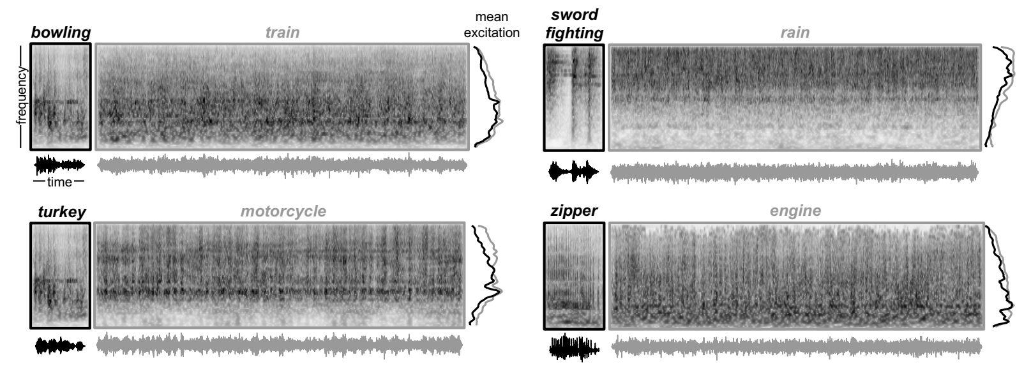
- Human listening abilities are remarkably robust to noise, but the computational basis of this robustness is poorly understood
- **We explored whether internal models of real-world noise—"noise schemas"—aid the perception of foreground sounds in noise**

Potential Explanations


Fixed Noise Suppression
stimulus
filters
filter responses
fixed filters attenuate noise-like sounds
Adaptive Noise Suppression
stimulus
filters
filter responses
responses to relatively constant features are suppressed over time via adaptation
Noise Modeling
stimulus
filters
filter responses
estimation of background noise statistics allows foreground sounds to be detected as outliers from the associated distribution
estimate distribution
evaluate likelihood

Stimuli

- Diverse set of 160 natural ***foreground sounds*** and ***background noises***
- ***Foreground sounds***: 0.5 s excerpts of recorded natural sounds
- ***Background noises***: 3.25 s excerpts of sound textures synthesized from statistics of real-world textures



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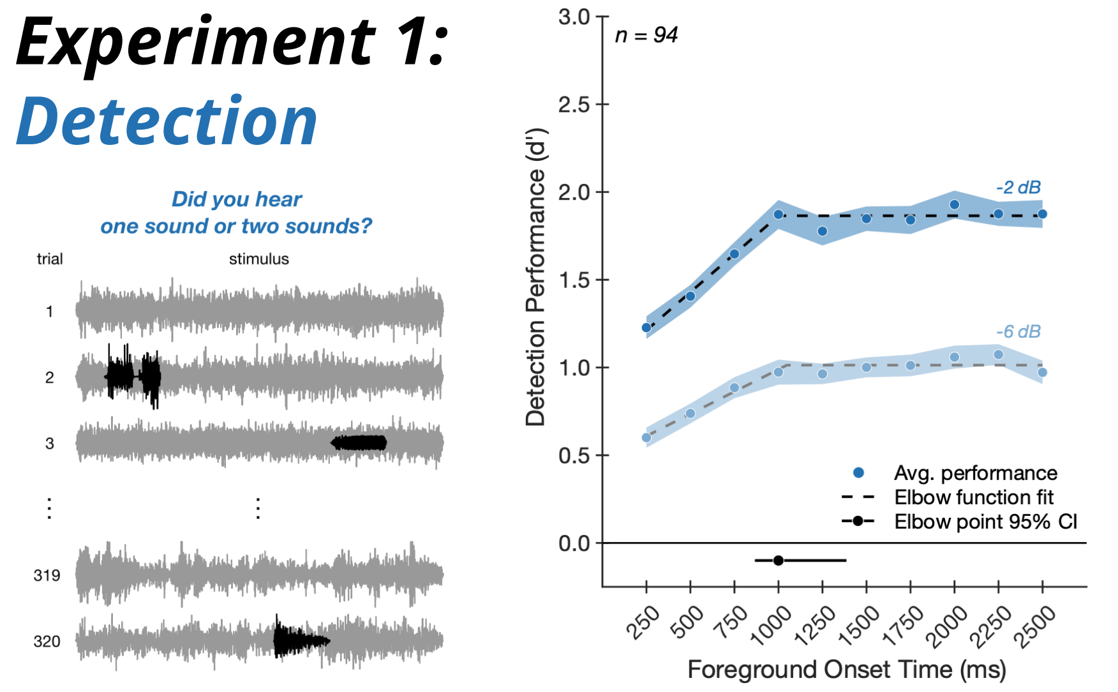
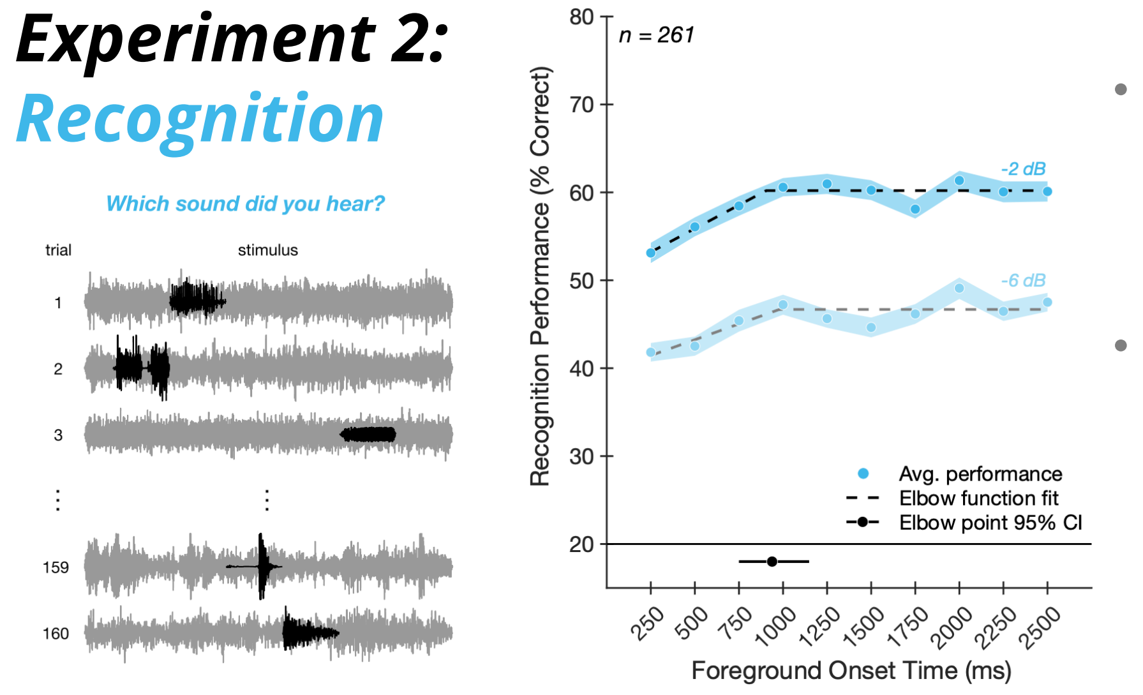
Experiment 1: *Detection*



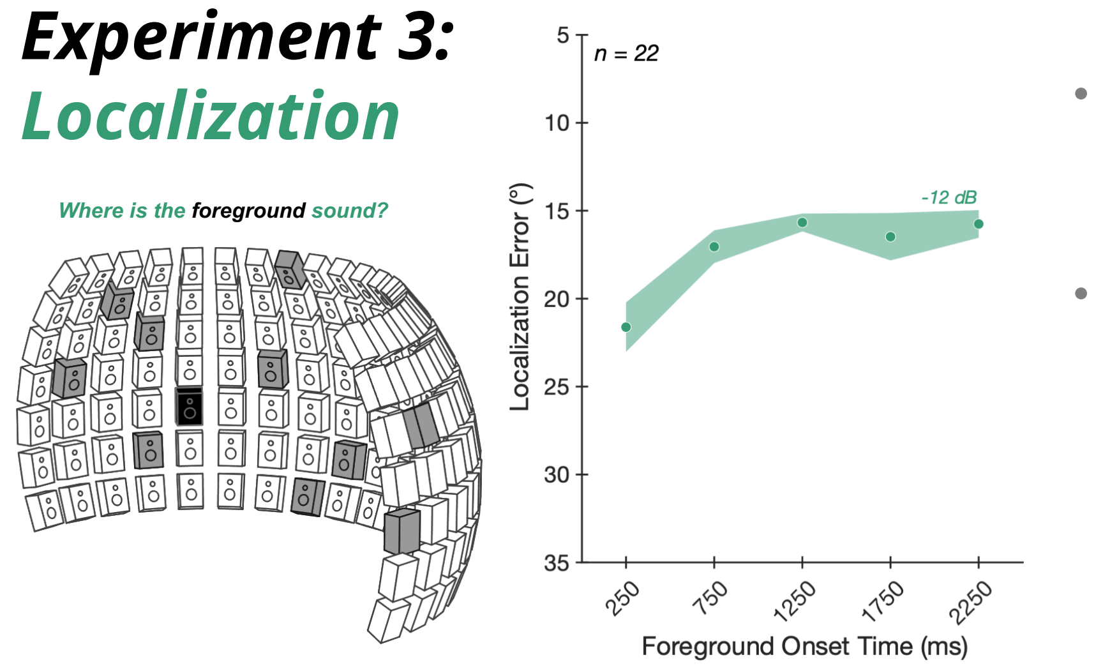
- Detection performance improved with exposure to the background, plateauing after one second of exposure to the background
- Benefit of background exposure was evident at both SNRs tested

Experiment 2: *Recognition*



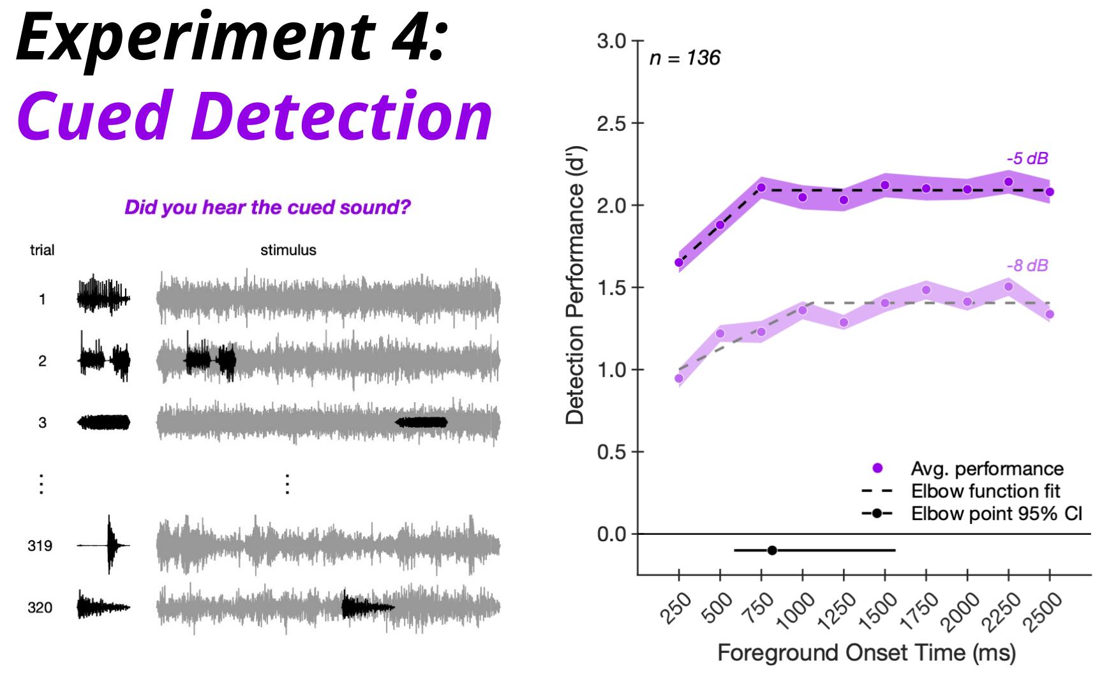
- Participants performed a recognition task on the stimuli from Experiment 1
- Benefit of background exposure extends to sound source recognition

Experiment 3: *Localization*



- Participants localized foreground sounds in diffuse background noise
- Foreground localization improved with exposure to the background like detection and recognition tasks

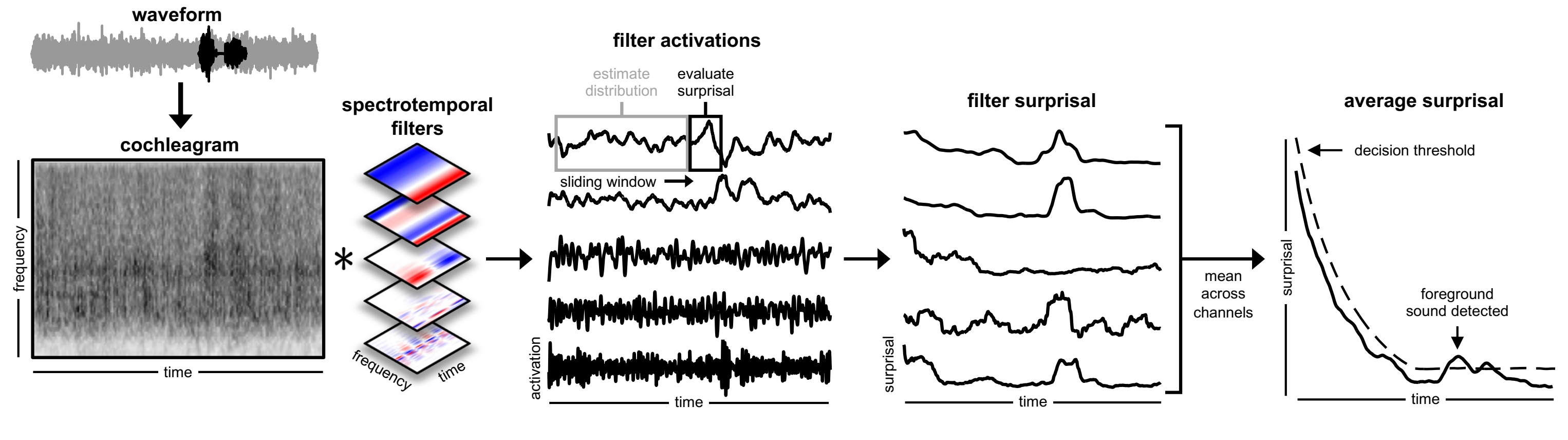
Experiment 4: *Cued Detection*



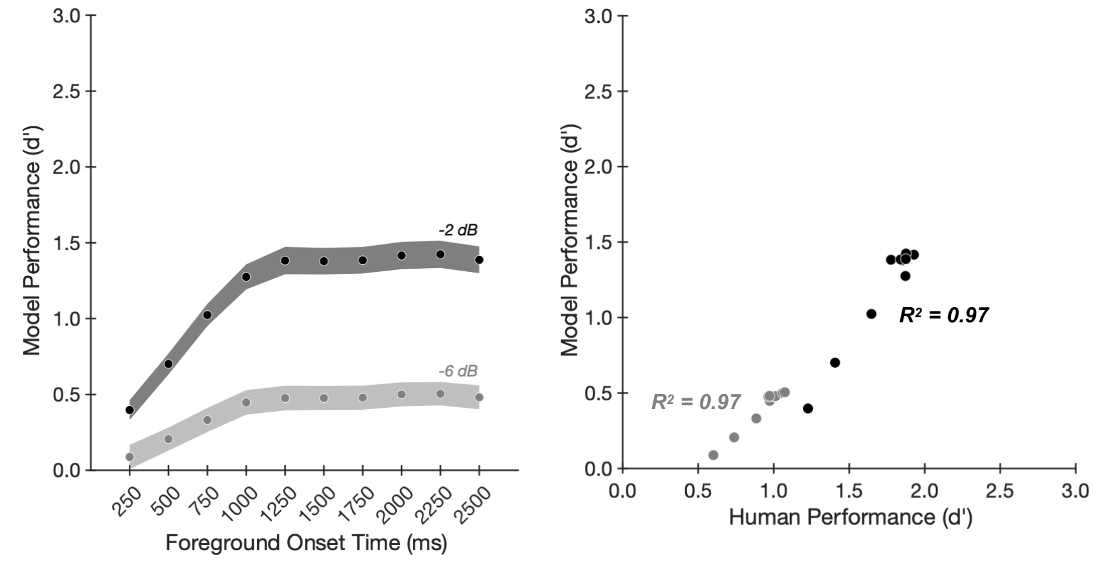
- Cued detection performance also improved with exposure to the background
- Benefit of background exposure persists even when participants know what to listen for

Modeling Results

- Model evaluates probability of samples under a distribution (univariate Gaussian) estimated from past samples
- Samples not belonging to background distribution will have low probability under a model of the background

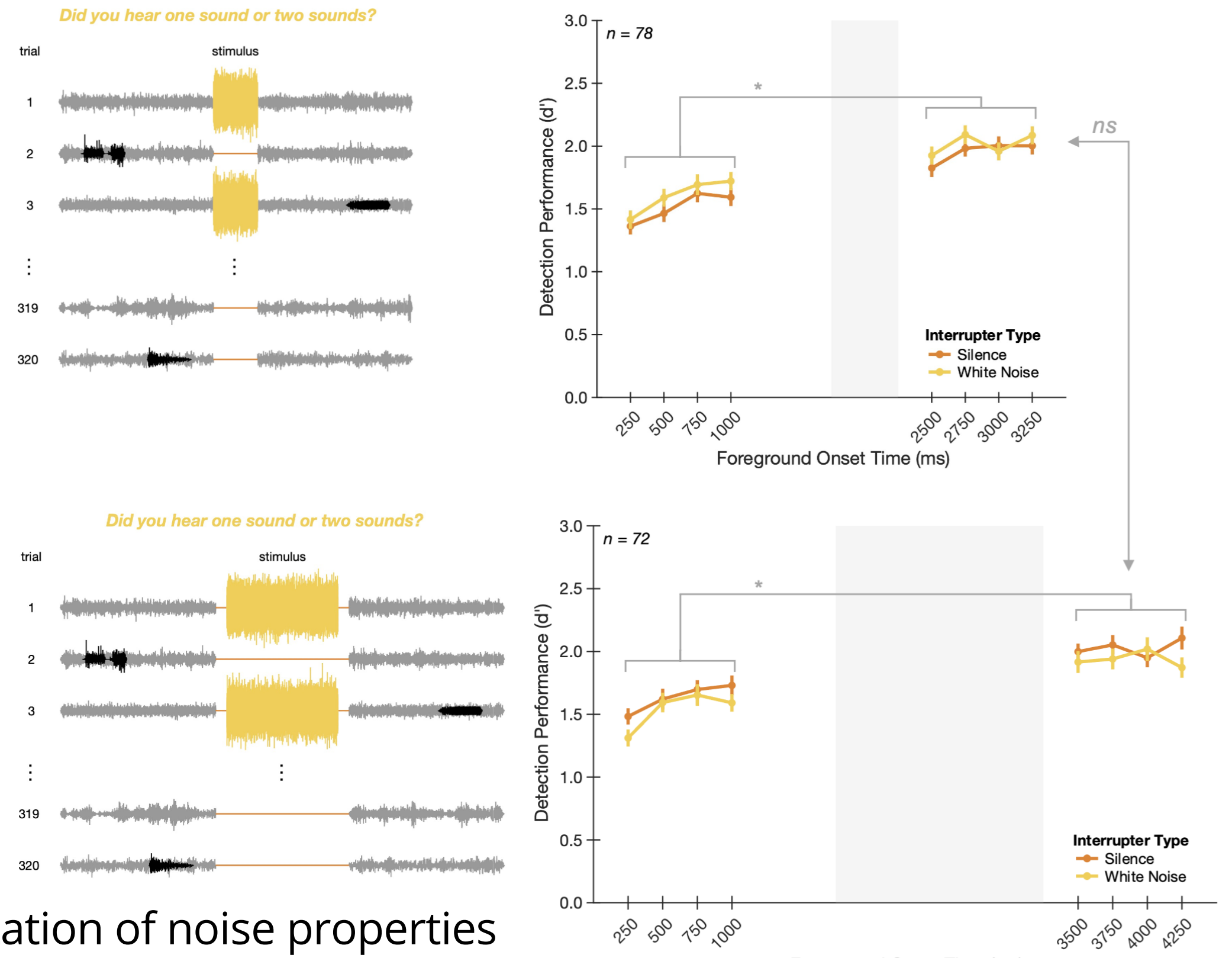


- Model replicated human results, showing a similar pattern of improvement with exposure to the background
- Human behavioral performance can be explained by a model that estimates the statistics of ongoing background noise



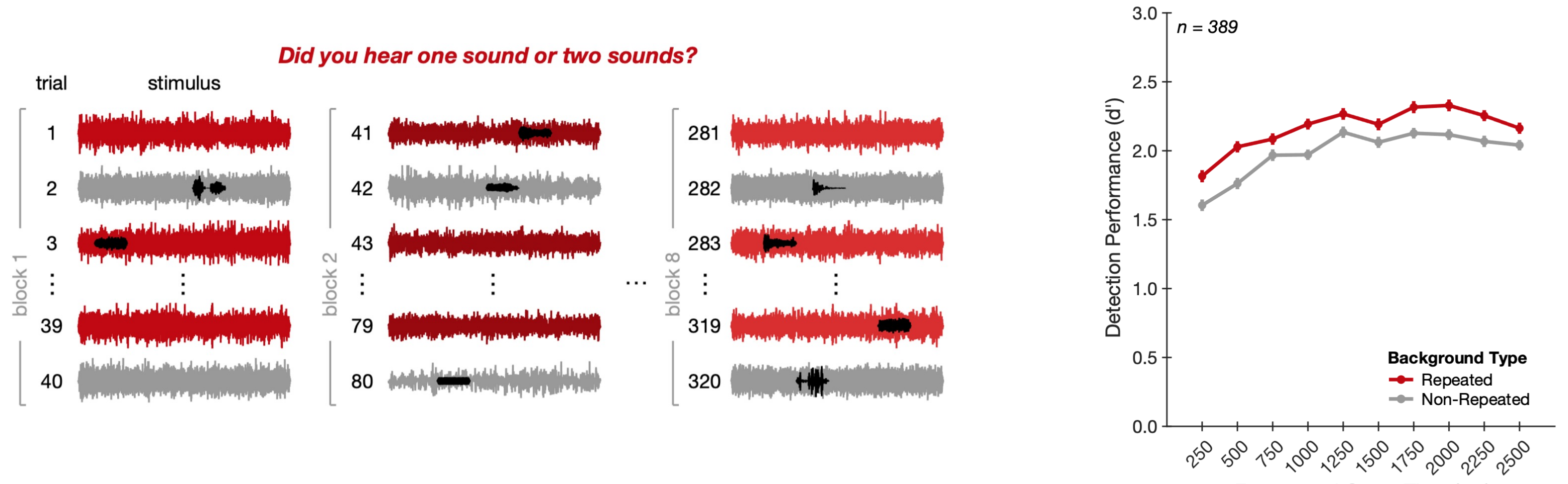
Experiments 5a & 5b: *Background Interruptions*

- Interrupted middle of background with either ***silence*** or ***white noise***
- Change to the background might cause adaptation to "reset", leading to a decrement in performance following the interruption
- The noise modeling hypothesis could allow for a benefit of background exposure, since estimated noise properties can be stored across the interruption
- Detection performance was greater following the interruption and was not affected by the duration of interruption
- Suggest listeners maintain representation of noise properties across temporary interruptions in background noise



Experiment 6: *Intermittent Repeated Backgrounds*

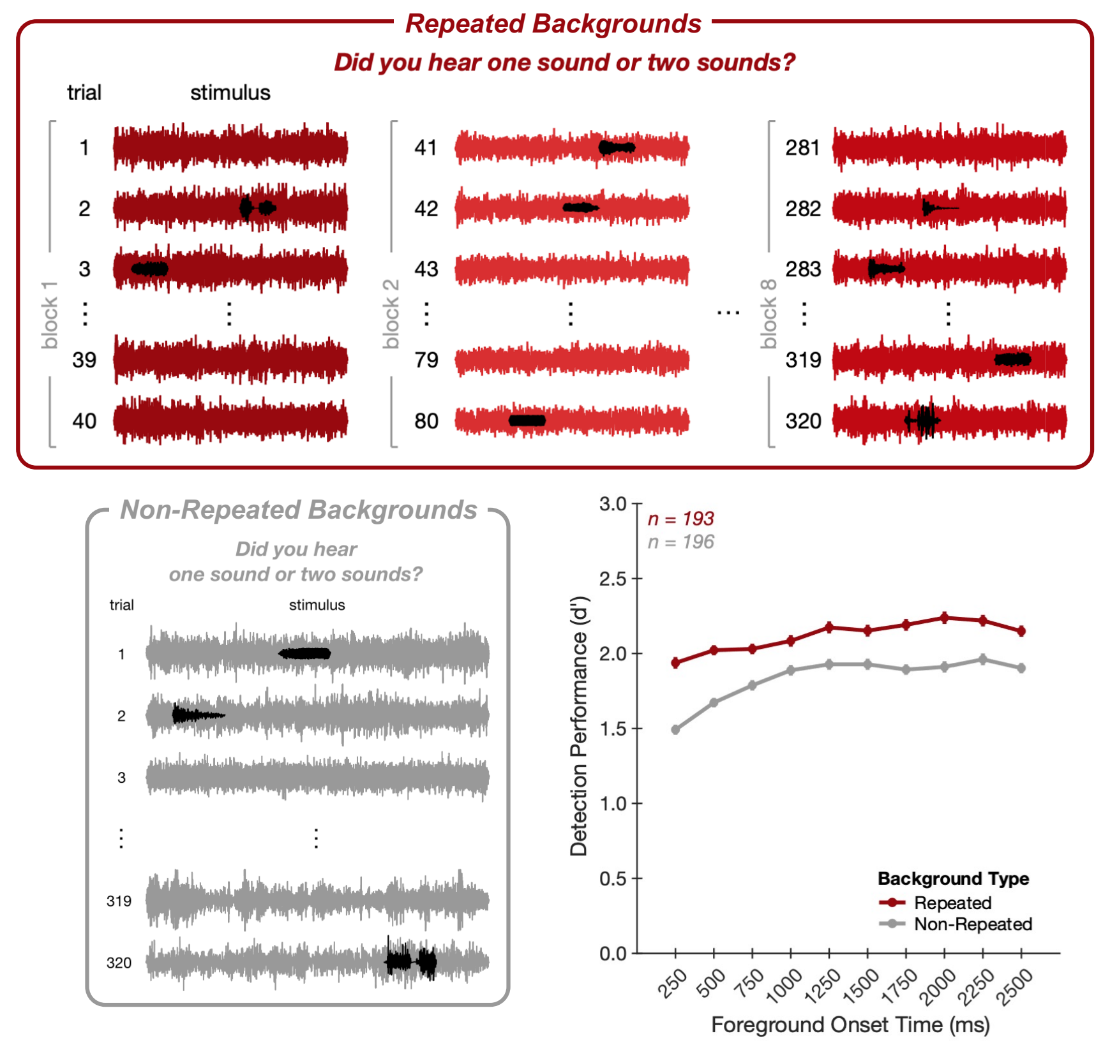
- Are internal models of background noise maintained over longer periods of time and across intervening stimuli, akin to "schemas" learned for patterns in speech and music?
- If listeners learn and use noise schemas to aid hearing in noise, detection performance should be enhanced for frequently recurring background noises



- Detection performance was enhanced for recurring background noises
- Suggests that noise schemas (representations of noise statistics that aid detection) are built up and maintained across intervening stimuli

Experiments 7 & 8: *Repeated Backgrounds on Every Trial*

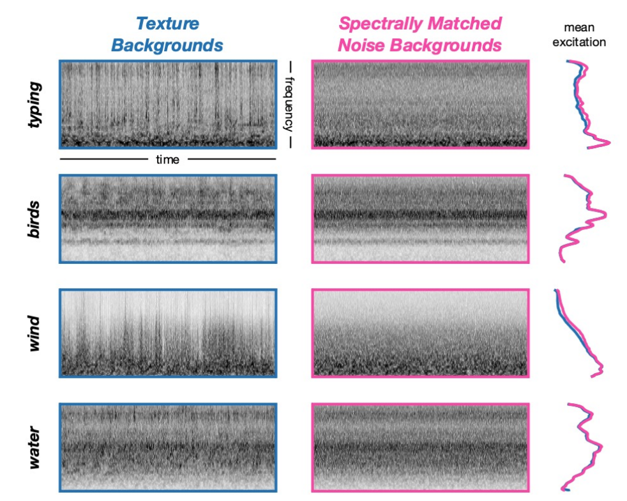
- If listeners use stored representations of noise statistics to aid detection, then effect of temporal position within noise should be reduced for repeated backgrounds

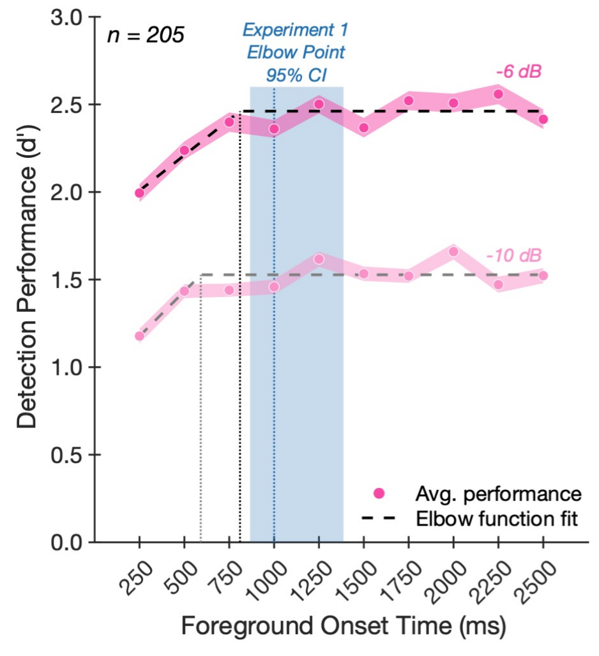


- The effect of onset time was smaller for the ***repeated*** vs. ***non-repeated*** backgrounds, suggesting that schemas learned for repeating backgrounds improve detection and lessen the temporal benefit of background exposure

Experiment 9: *Stationary Noise*

- If listeners estimate statistics within an integration window, then the window width might reflect a tradeoff between obtaining good estimates of noise statistics and resolving changes in statistics
- Since the accuracy of estimated statistics depends on the stability of the noise statistics (i.e., stationarity), the optimal window of estimation might be shorter for more stationary noise





- Both the timescale and magnitude of improvement with exposure to the background was reduced for more stationary noise
- Suggests that the timescale of integration depends on the background stationarity

Conclusions

- ***Detection***, ***recognition***, and ***localization*** in noise improved with exposure to the noise
- A ***model*** designed to detect outliers from a distribution of background noise accounted for this pattern of human performance
- Performance was robust to ***interruptions*** in the noise and enhanced for ***repeating*** noises, suggesting representations of noise properties are built up and maintained over time
- The timescale of improvement with exposure to the background depended on the noise ***stationarity***
- Overall, the results suggest noise robustness is supported by internal models—"noise schemas"—that capture the structure of noise and facilitate estimation of other concurrent sounds